



CIVITAS indicators

Share of unsuccessful deliveries (TRA_FR_EF2)

DOMAIN

 Transport	 Environment	 Energy	 Society	 Economy
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TOPIC

Freight

IMPACT

Operational efficiency of urban deliveries
Improved efficiency and reliability of last-mile delivery operations.

TRA_FR

Category

Key indicator	Supplementary indicator	State indicator
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CONTEXT AND RELEVANCE

Urban freight measures frequently aim to improve the reliability and efficiency of delivery operations while maintaining service quality. A reduction in unsuccessful deliveries indicates more efficient logistics organisation, improved delivery planning, better customer coordination, or the use of delivery concepts that increase first-attempt delivery success. These improvements can help reduce unnecessary vehicle kilometres associated with re-deliveries and contribute to lower congestion, emissions, noise, road safety risks, and pressure on scarce public space.

This indicator measures the share of delivery attempts that are unsuccessful within a defined geographical area and reference period. **It is a relevant indicator when the policy action aims to increase the efficiency of urban freight distribution. A successful action is reflected in a LOWER value of the indicator after the experiment compared to the BAU case.**

DESCRIPTION

This indicator measures the proportion of planned deliveries that cannot be successfully completed during the intended delivery attempt and therefore require a re-delivery, a return-to-depot or collection at an alternative location.

The indicator is expressed as **% of deliveries**.

METHOD OF CALCULATION AND INPUTS

The indicator should be calculated **exogenously** based on the required inputs and then coded in the supporting tool.

Method 1		
Survey delivery companies on a sample of delivery tours	Significance: 0.50	
INPUTS		
The following information is needed to compute the indicator:		
- The total number of delivery attempts made in the pilot area over a defined reference period by the logistics operators surveyed. - The number of unsuccessful delivery attempts over the same reference period.		
EQUATIONS		
The equation that should be applied to quantify the indicator is:		
$UDR = \frac{NotDelivered}{ADeliveries} \times 100$		
Where:		
<i>UDR</i> = Unsuccessful Delivery Rate		
<i>ADeliveries</i> = Total number of delivery attempts during the observation period		
<i>NotDelivered</i> = Total number of unsuccessful deliveries during the observation period		

ALTERNATIVE INDICATORS

This indicator measures the share of unsuccessful delivery attempts. Other indicators in the CIVITAS Evaluation Framework addressing the operational efficiency of urban deliveries include: **TRA_FR_EF1**, which measures the average distance travelled per delivery; **TRA_FR_EF3**, which measures the average time per delivery; and **TRA_FR_EF4** and **TRA_FR_EF5**, which measure the average load factor of delivery vehicles for B2B and B2C deliveries, respectively.